Subject: Manufacturing Processes (23A0340IT)

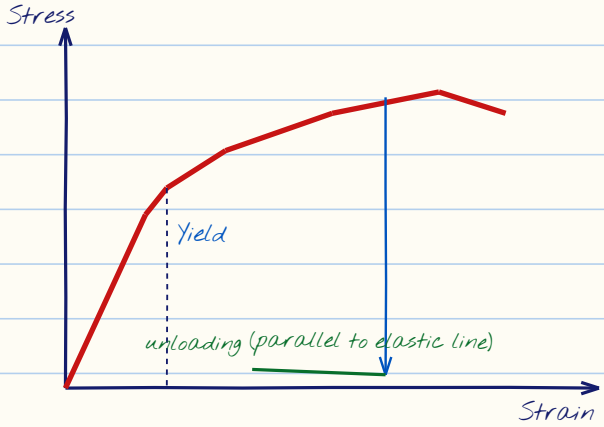
UNIT - III : BULK FORMING

Plastic deformation, recovery-recrystallization-grain growth, hot & cold working, strain hardening, annealing, forging, rolling, extrusion, wire & tube drawing

1. Plastic Deformation in Metals & Alloys

Bulk forming (metal working) changes the shape of a solid billet by plastic deformation - volume is conserved and the surface-area-to-volume ratio changes. Metal is stressed beyond its yield point.

- Elastic deformation: reversible, atoms displaced slightly; obeys Hooke's law ($\sigma = E \times \varepsilon$).
- Plastic deformation: permanent; occurs by slip (movement of dislocations along slip planes & directions) and twinning (mirror-image shear, mostly in HCP/BCC at low temp. & high strain rates).
- Real metals deform at much lower stress than theory because of dislocations; slip occurs on the close-packed planes (FCC - 12 slip systems -> most ductile; HCP - 3 -> less ductile; BCC - 48).
- Critical resolved shear stress: $\tau = \sigma \cos(\phi) \cos(\lambda)$ (Schmid's law) - slip begins when τ reaches τ_{crss}.
- True stress & strain: $\sigma_T = P/A_i$; $\varepsilon_T = \ln(L/L0)$; flow curve $\{r|\sigma = K \varepsilon^n\}$ (K = strength coefficient, n = strain hardening exponent).



Elastic: recoverable
Plastic: permanent (flow)
Strain hardening: stress rises as dislocations multiply & tangle
Necking: UTS (tension)

Fig 1.1 Stress-strain curve showing strain hardening

Strain hardening (work hardening)

As plastic strain increases, dislocation density rises (10^6 to 10^{12} $1/cm^2$); dislocations tangle and obstruct each other, so strength & hardness increase while ductility decreases. Cold working a metal: higher yield strength, hardness, brittleness, electrical resistivity increase; conductivity & corrosion resistance slightly fall; residual stresses and anisotropy appear.

2. Recovery, Recrystallization & Grain Growth

A cold-worked metal stores strain energy (about 10 % of the work). On heating (annealing) it moves back to the low-energy state in three overlapping stages:

Stage	What happens	Properties
Recovery (low temp, < 0.3 Tm)	Point defects annihilate; dislocations rearrange to low-energy cells (polygonisation); internal stresses relieved; no change of grain shape	Hardness / strength almost same; ductility slightly up; electrical conductivity recovers
Recrystallization (~0.3–0.5 Tm)	New strain-free equiaxed grains nucleate (at grain boundaries, slip bands) and grow, consuming deformed grains	Strength & hardness fall sharply; ductility restored; fully soft
Grain growth (higher T / longer time)	Large grains grow at the expense of small ones to reduce boundary energy	Strength falls further; coarse grain gives poor toughness & rough surface (orange peel)

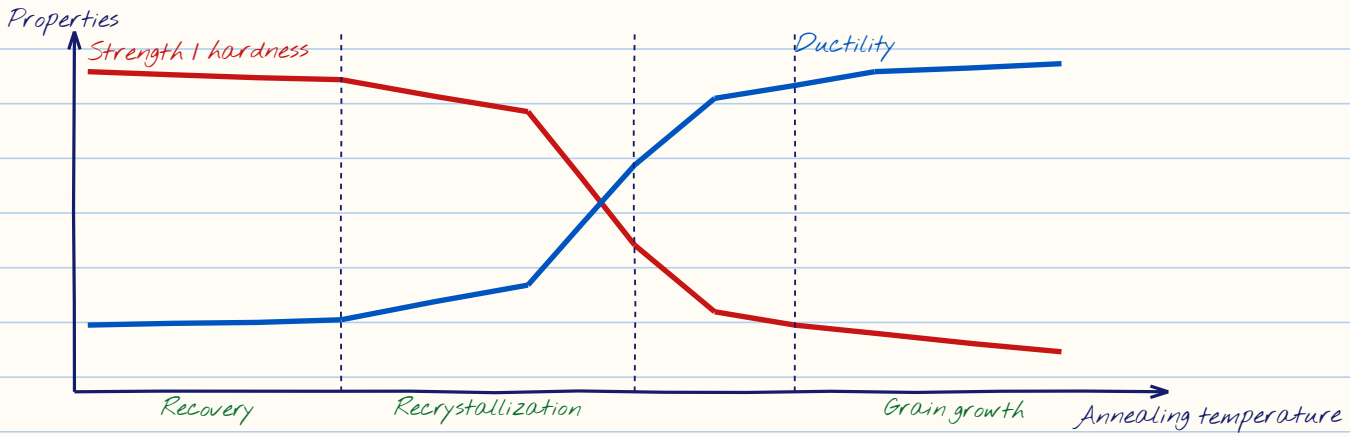


Fig 2.1 Effect of annealing on cold-worked metal

Recrystallization temperature (T_{rex})

- Lowest temperature at which complete recrystallization occurs in about 1 hour: about 0.3 to 0.5 of the absolute melting point (K). e.g. Fe ~ 450 C, Cu ~ 200 C, Al ~ 150 C, Pb & Sn - room temp.
- Lowers with: more prior cold work, finer initial grain, pure metals, longer time. Rises with alloying/impurities.
- Final grain size smaller if: more deformation, lower annealing temp, shorter time. Critical deformation (2-8 %) gives very coarse grains - avoid.

3. Hot Working & Cold Working

Aspect	Hot working	Cold working
Temperature	Above recrystallization temp (0.5-0.75 Tm)	Below recrystallization temp (room temp usually)
Force required	Low (flow stress is low)	High (strain hardening)
Strain hardening	None - recrystallizes continuously	Yes - strength, hardness up
Grain structure	Refined, equiaxed; blow holes closed; fibrous flow lines	Distorted, elongated grains; anisotropy
Surface finish & accuracy	Poor; scale formed, oxidation, tolerance +- 1-2 %	Good finish, close tolerance (+- 0.1 mm)
Ductility	High, large deformation in one step	Limited; intermediate annealing is needed
Residual stress	Nil	Present (cause distortion)
Applications	Forging, hot rolling, extrusion, hot piercing - ingots, large sections	Cold rolling sheet, wire drawing, cold heading, bolts
Cost	Heating & scale loss	Less energy but higher power & tooling

- Warm working: between the two (0.3-0.5 Tm) - reduces forces compared with cold, better finish than hot.
- Annealing: heating to a suitable temperature, holding, and slow furnace cooling to soften the metal, relieve stresses, refine grain, improve machinability & ductility. Types: full, process (recrystallization) annealing, stress-relief, spheroidising, normalising (air cooling).
- Hot-working effects: homogenises structure, breaks dendritic cast structure, heals porosity, refines grains, and creates fibre flow lines (mechanical fibering)

which give directional strength - e.g. crankshafts forged along the contour.

Effect of temperature & strain rate on flow stress

- Flow stress falls with rise in temperature; at hot-working temperature it is nearly independent of strain (no hardening) but depends on strain rate: sigma = C (strain rate)^m (m = strain-rate sensitivity: 0.02-0.4 hot, ~0.01 cold).
- High m (superplastic alloys, m > 0.5) resists necking -> huge elongations (500-2000 %) in superplastic forming.
- Hot shortness: some metals (S in steel) crack when hot worked - Mn added to form MnS. Cold shortness: brittleness at low temp due to P.

Solved example 1 - true strain

A bar 100 mm long is cold worked to 150 mm length. Engineering strain e = (150-100)/100 = 0.5; true strain epsilon = ln(150/100) = 0.405. If K = 500 MPa & n = 0.2, flow stress = 500 x $(0.405)^{0.2}$ = 500 x 0.834 = 417 MPa.

4. Forging

Forging is a bulk-forming process in which metal is shaped by compressive forces applied through hammers or presses (dies). Used for crankshafts, connecting rods, gears, turbine discs, spanners, bolts. Provides grain flow, high strength and toughness (better than castings).

Types of forging

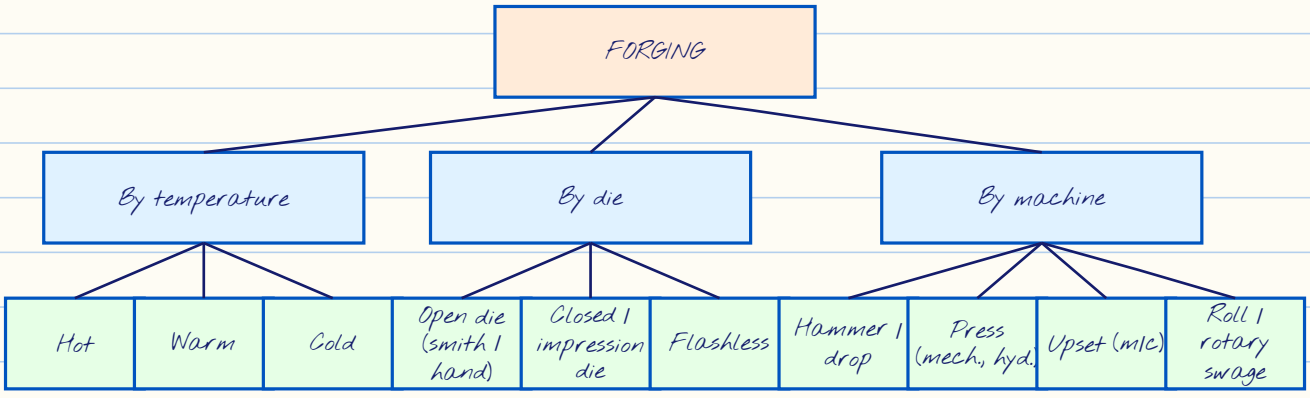


Fig 4.1 Classification of forging

Type	Description & uses
Open-die (smith) forging	Workpiece between flat/simple dies; metal flows freely. Operations: upsetting, drawing-out (cogging), fullering, edging, bending, punching, cutting. Heavy shafts, rings, discs.

Closed / impression-die (drop) forging	Heated billet in two shaped dies; excess metal flows out as flash (gutter) giving pressure to fill the cavity. Close tolerance, mass production: connecting rods, spanners, gears.
Upset forging	Horizontal forging machine; a bar is gripped and its end is upset (head formation): bolts, rivets, valve stems; cold heading.
Press forging	Slow squeezing action (hydraulic: 100–50000 t; mechanical: crank) – deep penetration, uniform flow, dies with less draft.
Roll forging	Bar passed through a pair of grooved segment rolls to reduce areal taper: axles, tapered levers, leaf springs.
Swaging / rotary	Dies hammer a rod rapidly while rotating – reduces diameter or makes tapers (tapered pins, tube ends).
Flashless / precision	Closed die with accurate billet volume; no flash; near net shape.

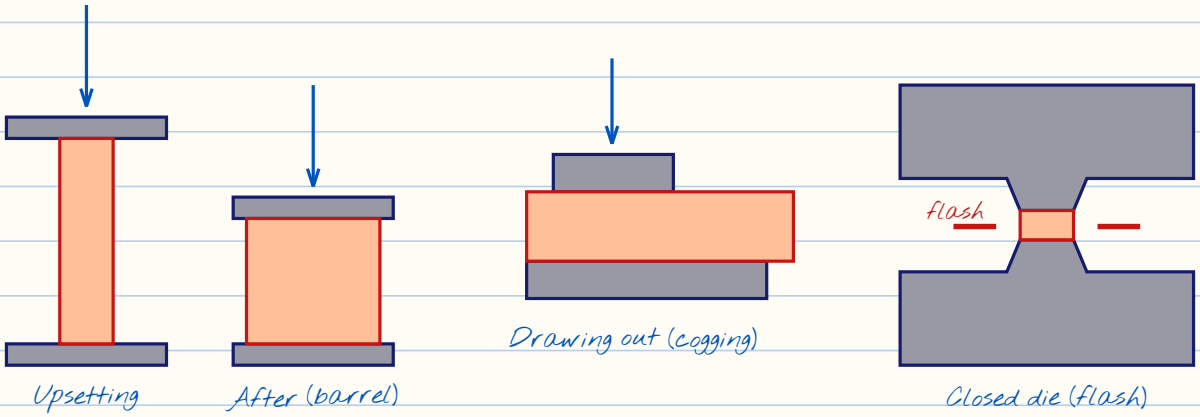


Fig 4.2 Forging operations

Steps in a typical drop-forging cycle

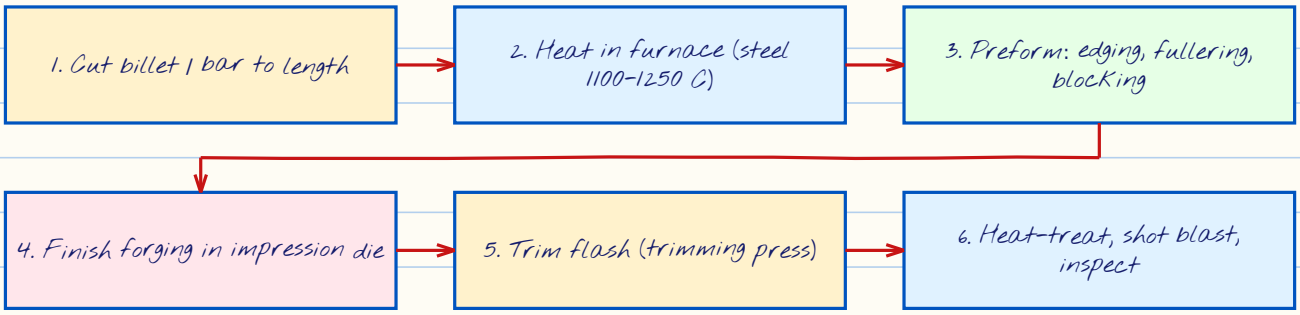


Fig 4.3 Forging sequence

- Heating: gas / oil-fired, induction or resistance heating; temperature ranges – Al 400–480 C, Cu alloys 700–900 C, carbon steel 1000–1250 C, Ti 800–1000 C. Overheating causes grain growth / burning; scale loss 1–3 %.
- Lubricants: graphite in oil / water, glass, MoS_2 – reduce friction & die wear and help release.

- Advantages of forging: high strength-to-weight, grain flow follows contour, no internal voids, good fatigue & impact strength, high reliability, good surface (cold forging). Limitations: high tooling cost, flash waste, limited to simple shapes without undercuts, scale & distortion in hot forging, poor tolerance.

Solved example 2 - upset forging load

A cylinder $d = 100$ mm, $h = 50$ mm is upset between flat dies; flow stress $\sigma_f = 200$ MPa, $\mu = 0.2$. Area $A = \pi/4 \times 100^2 = 7854$ mm^2. $F = \sigma_f A (1 + \mu d / 3h) = 200 \times 7854 \times (1 + 0.2 \times 100 / 150) = 1.5708 \times 10^6 \times 1.133 =$ 1.78 MN (about 178 tonnes).

Forging equipment & force

- Hammers: board drop (gravity), power (steam / air) - impact; energy $E = 0.5\, m\, v^2$. Presses: mechanical (crank / knuckle) - stroke limited; hydraulic - constant force.
- Forging load (open-die upsetting of cylinder): $F = \sigma_f \times A \times (1 + \mu d / 3h)$ (σ_f = flow stress, A = contact area, d = diameter, h = height, μ = friction coefficient). Friction increases pressure towards centre - friction hill.
- Closed-die load roughly $F = K \sigma_f A$, with $K = 3$-12 depending on shape complexity.
- Die design: parting line, draft (3-7 degrees), fillets & corner radii, flash gutter, material allowance, shrink allowance (about 1.2 %).

Forging defects & remedies

Defect	Cause	Remedy
Laps / folds (cold shut)	Metal folded over and pressed in without fusing; sharp die corners; wrong billet size	Generous fillets, correct preform and material flow
Incomplete die fill / unfilled section	Insufficient metal, cold die/billet, poor lubrication, low pressure	Proper billet volume & temp, adequate force, preheated die
Surface cracks (hot shortness)	Working below proper temp or sulphur-rich steel; excessive reduction	Heat to correct range, slow deformation, control S
Scale pits	Oxide scale pressed into surface	Descale before forging, protective atmosphere, lubricants
Die shift / mismatch	Misalignment of upper & lower dies	Align dies, lock with dowels / guides

Buckling	Long slender billet in upset (h/d > 3)	Limit L/d < 3, guide dies
Burning / overheating	Very high heating (coarse grain, brittle)	Control furnace temp
Flakes / internal bursts	Hydrogen & cooling stress	Slow cooling, anneal

5. Rolling

Rolling reduces thickness (or changes cross-section) of a long workpiece by compressive forces from two (or more) rotating rolls. It is the most widely used metal-working process (~90 % of all metal is rolled at least once). Hot rolling of ingots/billets -> blooms, billets, slabs; then finish rolling.

- Products: bloom (> 150 x 150 mm sq.), billet (40–150 mm sq.), slab (rectangular, > 200 mm wide), plates (> 6 mm), sheets (< 6 mm), strips, foils, rods, bars, rails, structural shapes (I, H, channels, angles), tubes (Mannesmann piercing), rings, threads.
- Flat rolling parameters: initial thickness h0, final hf; draft d = h0 − hf; roll radius R; contact length L = sqrt(R (h0 − hf)); reduction r = d/h0; strain epsilon = ln(h0/hf).
- Bite condition: roll grips if tan alpha <= mu; max draft = mu^2 R (alpha = angle of contact = sqrt(d/R)).
- Neutral (no-slip) point: where roll velocity = strip velocity; before it strip slower than roll (friction pulls in), after it faster (forward slip).
- Roll force F = L x w x sigma_avg (w = width, sigma_avg = average flow stress in the bite). Torque T = F x (L/2) per roll (arm approximately 0.5 L); Power P = 2 pi N T x 2 rolls = 2 x 2 pi N F L/2 W = 2 pi N F L (N in rev/s).

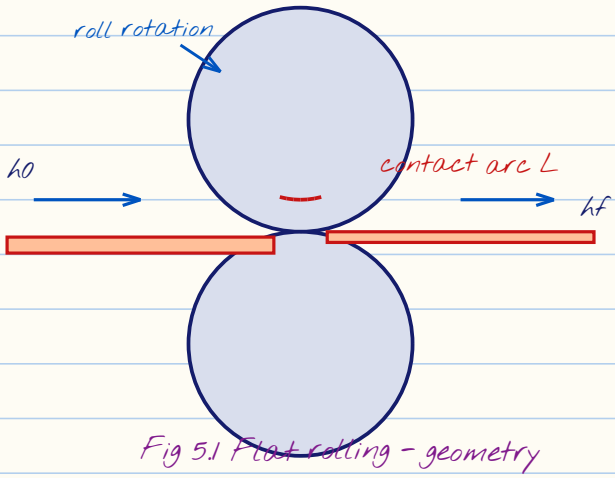


Fig 5.1 Flat rolling - geometry

Neutral point: v(roll) = v(strip)

Entry: strip slower than roll

Exit: strip faster (forward slip)

F = L w sigma_avg

Power = 2 pi N F L

Types of rolling mills

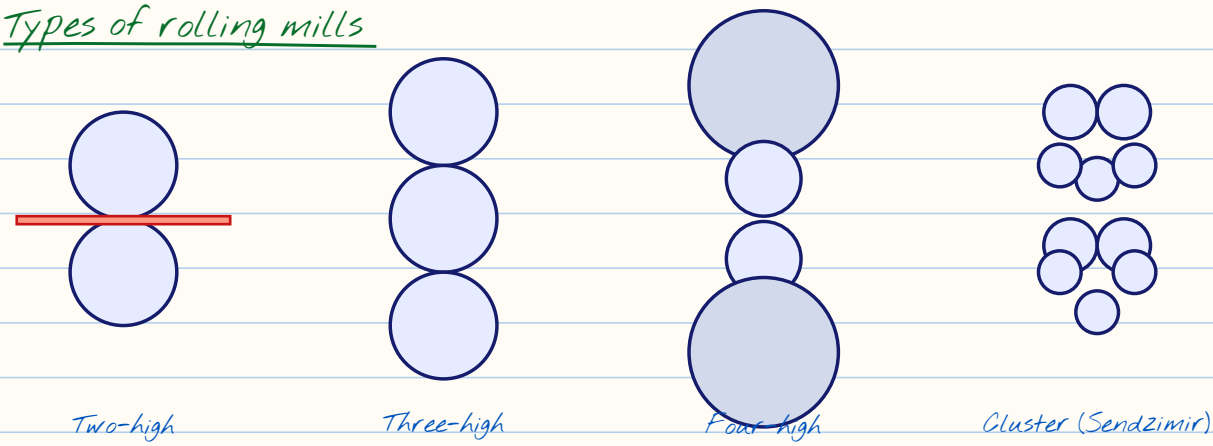


Fig 5.2 Rolling mill arrangements

Mill	Features & use
Two-high (reversing / non-reversing)	Simple; two rolls same size. Reversing - hot rolling blooming, slabbing, plates, rails. Non-reversing - continuous finishing.
Three-high	Upper, middle, lower rolls; the work passes through between alternate pairs; no reversal needed; for structural sections, rails (billets).
Four-high	Two small work rolls backed by two large backup rolls -> reduces roll deflection & force; for plates, sheets, strip (hot & cold).
Cluster (Sendzimir)	Small work rolls supported by many backup rolls - thin foils / hard stainless strip.
Tandem (continuous)	Series of stands, strip simultaneously in several - high speed strip production (hot strip mill / cold mill).
Planetary & ring rolling	Two heavy backup rolls with many small planetary rolls - 90 % reduction in one pass; ring mills for bearing races, wheel rims.

Hot rolling	Cold rolling
Done above T_rex; ingot -> bloom / slab / billet; large reductions (20-50 % per pass)	Done at room temp on hot-rolled, pickled strip; small reductions (< 30 %)
Rough surface, scale, tolerance 2-5 %; thickness above 2 mm	Bright finish, close tolerance (+- 0.01 mm), thin sheets, foils
No work hardening; low roll force	Work hardened (temper-rolled); higher force; roll flattening
Products: plates, rails, I-beams, bars	Products: car-body sheet, tinplate, strip, foils

Seamless tube production (Mannesmann piercing)

- Shaped products: roll-pass design - bar passes through successive grooves (box,

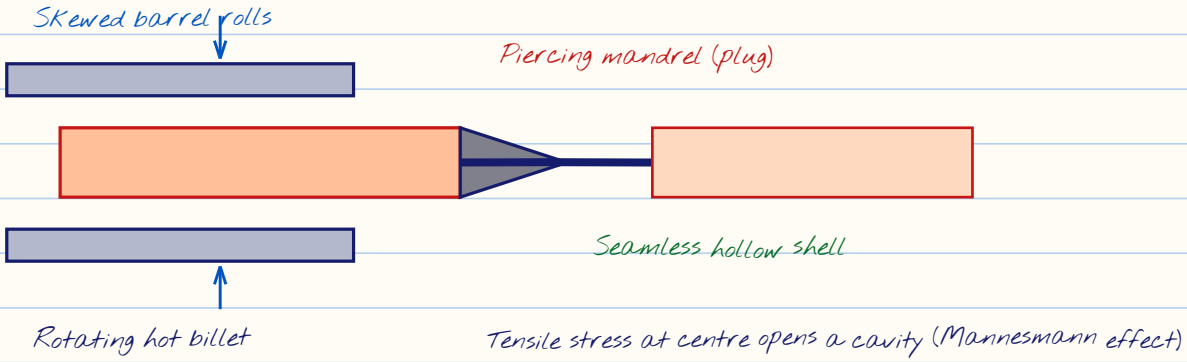


Fig 5.3 Rotary piercing of a hot billet

diamond, oval, round) to give rods / angles / rails; spread is the increase in width (0.35 x draft).

- Rolling of rings, threads (cold, thread rolling dies - 10x faster than cutting, stronger), gears, shot peening.
- Controls: roll gap, roll speed, tension (front / back), lubrication (cold - emulsion), roll cooling water.

Solved example 3 - flat rolling

Plate 25 mm thick is rolled to 20 mm with rolls of $R = 250$ mm, width $w = 300$ mm, $\sigma_{avg} = 250$ MPa, $N = 100$ rpm, $\mu = 0.1$.

- Draft = 5 mm; max draft = $\mu^2 R = 0.01 \times 250 = 2.5$ mm per pass => cannot be done in one pass by friction alone; two passes required (or lubricant removed).
- Contact length $L = \sqrt{250 \times 5} = 35.4$ mm. Roll force $F = L w \sigma = 35.4 \times 300 \times 250 = 2.65$ MN.
- Power $P = 2\pi N F L = 2\pi (1.667)(2.65 \times 10^6)(0.0354) =$ about 982 KW.

Rolling defects

- Wavy edges: rolls bend (deflection) -> edges elongate more; cure - roll crowning, back-up rolls, bending.
- Zipper / centre cracks, edge cracks: low ductility, excessive reduction; Alligatoring: ingot splits in two tongues (non-uniform deformation); Residual stress / camber, scale, laps.

6. Extrusion

Extrusion forces a billet through a die opening by a ram, producing a long section of constant cross-section: rods, tubes, I/T/L shapes, window frames, aluminium profiles. Extrusion ratio $R = A_0 / A_f$ (typically 10-100). Billet is hot (Al: 400-500 C; Cu: 700-900 C; steel 1100-1250 C) or cold, with lubricants (glass, graphite).

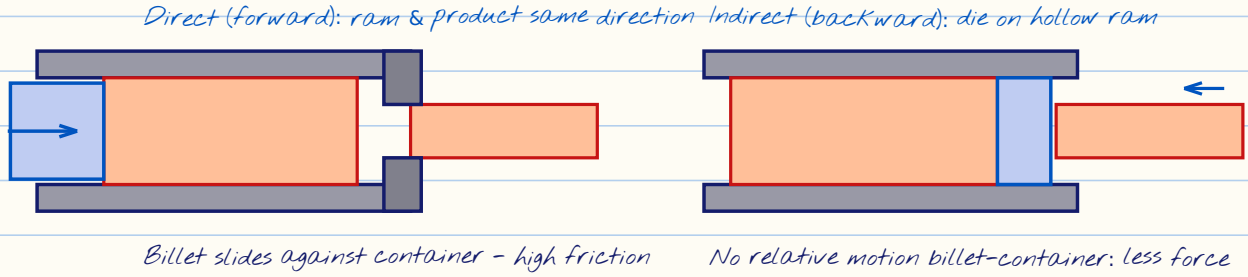


Fig 6.1 Direct and indirect extrusion

- Direct: most common; ram pushes billet through die at the opposite end; dummy block; friction with container -> higher force (+25-30 %), butt remains (10 % scrap).
- Indirect (backward): die at the ram end moves; product comes out through the hollow ram; billet static - friction absent, 25-30 % lower force; but ram support limits length.
- Hydrostatic: billet surrounded by pressurised fluid (castor oil, 1400 MPa) - no billet-container friction; brittle materials (Mo, W, ceramics) extruded; high reduction (up to 100:1); costly. Clad wires also.
- Impact extrusion: cold extrusion with a fast punch hitting slug placed in die: product moves forwards|backwards|combined - thin-walled tubes: toothpaste tubes, aluminium cans, battery cases, aerosol containers; Al, Pb, Sn, Cu, Mg slugs.
- Tube extrusion: mandrel with the ram forms hollow shapes; seamless tubes also by piercing + extrusion.
- Extrusion force: $F = A0 \times K \times \ln(A0/Af)$ where K = extrusion constant (MPa; Al 250 hot, steel 600+) (plus friction & redundant work terms).

Solved example 4 - extrusion

Billet dia 100 mm is extruded to a round rod 40 mm dia (Al, K = 250 MPa). Extrusion ratio $R = (100/40)^2 = 6.25$; $\ln R = 1.833$; $A0 = 7854\ mm^2$. Ram force $F = A0\ K \ln R = 7854 \times 250 \times 1.833 = 3.6\ MN$ (360 t). Die angle 90 degrees (flat) or conical (30-60) reduces dead-metal zone.

- Dies: flat (square) die - dead metal zone, good surface; conical|streamlined - lower force; porthole | bridge dies for hollow sections (aluminium window frames); die steel H13, tungsten carbide.
- Lubrication: hot - glass powder, graphite (Ugine-Sejournet for steel); cold -

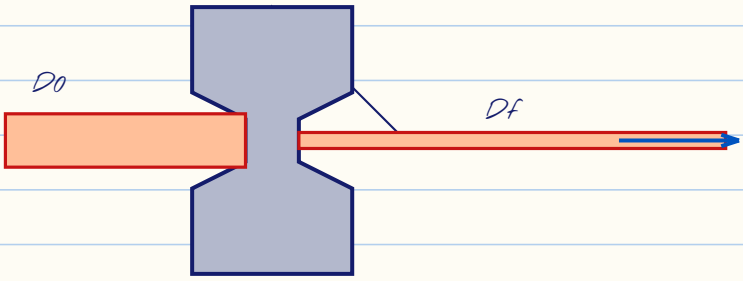
phosphate coat + soap.

Extrusion characteristics, defects

- Advantages: complex cross-sections, high reduction in one step, good surface & dimensional accuracy, no waste; seamless tubes; compressive stress state allows brittle metals.
- Defects: centre-burst (chevron cracking) due to tensile stress at centre (small die angle, low reduction); piping (tail-pipe) - surface oxide drawn to centre at the end; surface cracks (fir-tree) - speed/temperature too high; also die wear, shape distortion.

7. Wire Drawing & Tube Drawing

Drawing reduces the cross-section of a rod/wire/tube by pulling it through a tapered die. Unlike extrusion, material is pulled (tensile at exit) and compressed laterally. Mostly cold drawing - excellent finish, tolerance and strength. Wire diameter 0.025-12 mm; drawing speed up to 50 m/s.



Die: approach (alpha 6-10 degrees), bearing land, back relief; WC / diamond

Draw stress = sigma_avg ln(A0/Af)

Reduction per pass 20-40 %

Annealing between passes

Fig 7.1 Wire drawing through a die

- Drawing stress sigma_d = sigma_avg ln(A0/Af) (ideal, frictionless, homogeneous); with friction & redundant work about 1.2-2x. Maximum possible reduction per pass: about 63 % (when sigma_d equals flow stress) - practical 20-50 %.
- Process: pointing, pickling (descaling), coating (lime, phosphate, borax), lubrication (soap powder / oil), draw, repeated multi-die (continuous, tandem) with intermediate annealing, final stress relief.
- Die materials: tungsten carbide (steel), diamond (fine wires), tool steel. Lubricants: soaps, grease, oils.
- Defects: centre-burst, seams, die-marks, surface scratches, breaks; wire cold-worked becomes harder: tensile strength up to 2500 MPa (piano wire).

Tube drawing

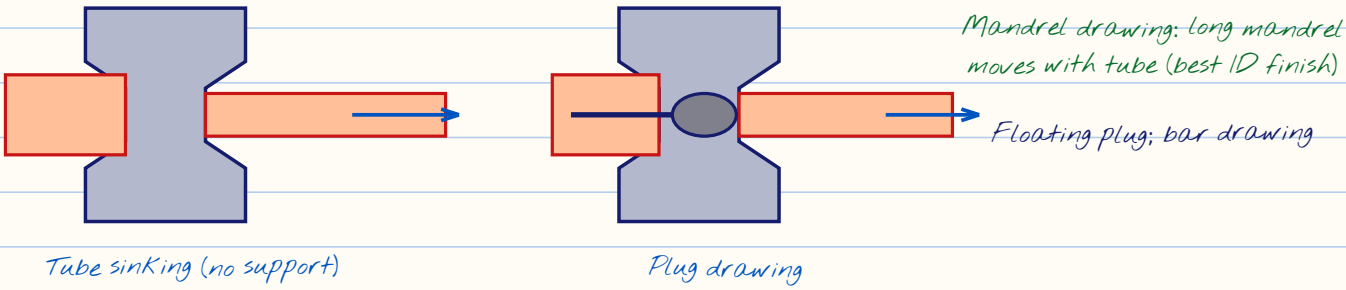


Fig 7.2 Tube drawing methods

- Tube sinking: only OD reduced; wall thickens; ID rough. Plug drawing: fixed/floating plug controls ID & thickness. Mandrel drawing: long mandrel supports the tube; draws with the tube - accurate ID, for long tubes.
- Used for seamless pipes, hypodermic needles, heat-exchanger tubes, bicycle frames; hot-extruded/pierced tubes are then cold drawn for accuracy.

Solved example 5 - wire drawing

A 10 mm wire is drawn to 8 mm; sigma_avg = 300 MPa. Area ratio A0/Af = $(10/8)^2$ = 1.5625; ln = 0.446; reduction = 1 - 0.64 = 36 %. Ideal draw stress = 300 x 0.446 = 134 MPa (< flow stress of exit wire, so wire will not break). Max reduction (ideal) = 1 - 1/e = 63 %.

Formula sheet - Unit III

Process	Formula	Terms
True strain / stress	epsilon = ln(L/L0); sigma_T = sigma_e (1 + e)	e = engineering strain
Flow curve	sigma = K epsilon^n	n = strain-hardening exponent
Recrystallization	T_rex = 0.3 to 0.5 Tm (K)	Tm = melting temperature
Upset forging	F = sigma_f A (1 + mu d / 3h)	A = contact area
Rolling	L = sqrt(R dh); F = L w sigma; P = 2 pi N F L; max dh = mu^2 R	dh = draft, N = revs
Extrusion	R = A0/Af; F = A0 K ln R	K = extrusion constant
Drawing	sigma_d = sigma_avg ln(A0/Af); max reduction = 63 %	ideal, no friction

Quick viva / 2-mark questions

- Define strain hardening. -> Increase in strength & hardness due to plastic deformation (dislocation pile-up).
- Why is hot working done above T_rex? -> To get continuous recrystallization, no hardening, low force.
- What is flash in forging? -> Excess metal squeezed out between dies - helps fill cavity & is trimmed.
- Why back-up rolls? -> To prevent deflection of small work rolls. What is the neutral point in rolling?
- What is extrusion ratio? -> A0/Af. What is piping? -> Oxide & impurities drawn in centre, tail-end defect.
- Difference between wire drawing & extrusion: pulling vs pushing; tensile vs compressive.

Important exam questions - Unit III

- Explain recovery, recrystallization & grain growth. Compare hot working & cold working. (10 M)
- Strain hardening and annealing. (5 M)
- Types of forging; forging defects & remedies. (10 M)
- Rolling: principle, types of rolling mills, roll force & power derivation, defects. (10 M)
- Types of extrusion; impact & hydrostatic extrusion; extrusion defects. (10 M)
- Wire drawing and tube drawing with neat sketches. (10 M)